Magneto-optical Kerr microscopy investigation of magnetization reversal in Co_2FeSi Heusler alloy thin films

Based on the article by B. K. Hazra et al.

Introduction & Motivation

Why Co_2FeSi Heusler Alloy?

- Theoretical 100% spin polarization at the Fermi level (Half-metal).
- High Curie temperature making it thermally stable.
- Crucial for designing multilayer spintronic devices.

Objective: Understand magnetic anisotropy (MA) and magnetization reversal (MR) processes to control preferred crystallographic directions and device efficiency.

Sample Preparation:

- DC magnetron sputtering (UHV) on Si(111).
- **TS-series:** Fixed thickness (50 nm), variable temperature (RT to $600^\circ C$).
- **t-series:** Variable thickness (12-100 nm) at $550^\circ C$.

Static Magnetic Hysteresis Loops (L-MOKE)

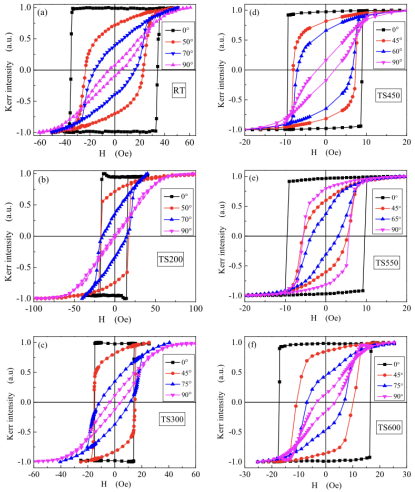

FIG 1: Static hysteresis loops at selected angles.

Key Observations:

- Rectangular hysteresis loops when the field is applied along the easy axis ($\varphi_H = 0°$).
- Normalized squareness ratio (M_r/M_s) is ≈ 1 along the easy axis.
- Maximum coercive field (H_c) observed along the easy axis.

Angular Dependence of Remanent Magnetization

Anisotropy Analysis:

- Polar plots show clear two-fold symmetry for most films.
- Maxima at 0° and 180° (easy axes), minima at 90° and 270° (hard axes).
- Exception: TS550 shows anomalous 4-fold symmetry due to two mutually exclusive uniaxial anisotropies.

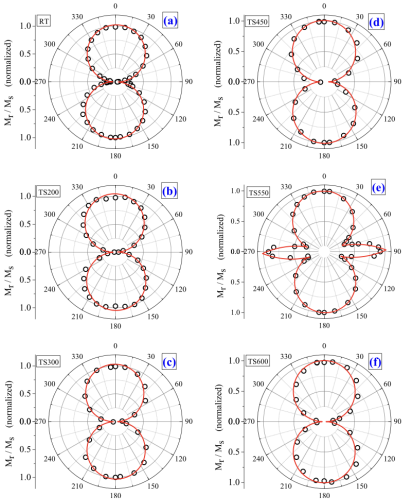

FIG 2: Polar plots of normalized remanent

Angular Dependence of Coercive Field

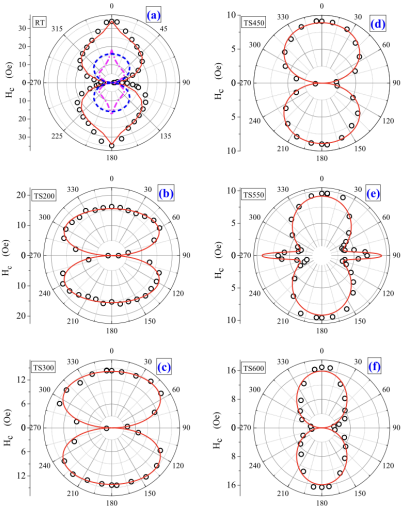

FIG 3: Polar plots of the coercive field (H_c).

Theoretical Fits:

- **Amorphous film (RT):** Reversal described by a mix of Stoner-Wohlfarth (SW) and Two-Phase (TP) models.
- **Crystalline films:** Completely and accurately described by the Two-Phase (TP) pinning model.

Direct Observation: Domain Dynamics

Magnetization Reversal Mechanisms:

- **Along Easy Axis (0°):** Nucleation of reverse domains and subsequent growth by rapid domain wall movement.
- **At 45°:** Reversal takes place through the formation of ripple domains.
- **Along Hard Axis (90°):** Field-induced movement of the 180° domain walls.

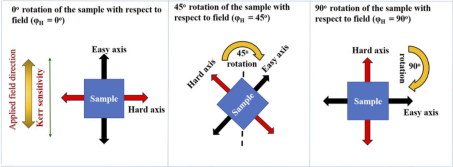

FIG 4: Kerr microscopy images of domain evolution.

Conclusions

- Co_2FeSi films exhibit strong in-plane uniaxial anisotropy (UA) driven by a "self-shadowing" effect during deposition.
- Uniaxial anisotropy persists regardless of amorphous or crystalline structural states.
- Reversal mechanism shifts from a mixed state (coherent rotation + pinning) in RT films to pure **domain wall pinning** in higher temperature films.
- The condition $H_c < H_k$ holds good for all samples, consistent with the pinning model.